

AirSense AI

Usability Testing & Analysis

Internship Task 4 — Final Submission

A usability evaluation of the AirSense AI digital product, including test planning, participant sessions, observations, metrics, findings, and design recommendations.

1. Usability Test Objective

The objective is to evaluate whether users can quickly understand current air quality, find health guidance, view the forecast, configure an alert, and navigate back to Home without assistance.

Product under test: AirSense AI mobile interface from the previous internship tasks.

Method: moderated task-based usability testing with think-aloud observation.

Sample: 5 demo participants representing students and a young working professional.

2. Test Scenarios

1. Find the current AQI.
2. Decide whether outdoor activity is recommended.
3. Find today's health advice.
4. Check the short-term forecast.
5. Set an alert for poor air quality.
6. Return to Home.

3. Participants & Metrics

Participant	Profile	Tasks	Success	Avg. time	Rating /5
P01	Student	5/5	92%	58s	4.5
P02	Student	5/5	88%	64s	4.0
P03	Student	4/5	76%	79s	3.5
P04	Professional	5/5	96%	51s	4.5
P05	Student	5/5	90%	61s	4.0

Key metrics: task completion, task success rate, average task time, assistance required, errors, hesitation points, and post-test satisfaction rating.

Demo aggregate: 88.4% average task success, approximately 63 seconds average task time, and 4.1/5 average satisfaction.

Important: These participant figures are illustrative/demo data created for the internship deliverable; they are not claimed as real research.

4. Observations

- Most users immediately recognized the AQI card as the main status indicator.
- Users understood the advice once they reached the Health Advice screen.
- One participant hesitated because the relationship between AQI severity and recommended action was not explicit enough.
- The Forecast → Set Alert path was understandable, but an alert shortcut on Home would reduce navigation effort.
- Participants preferred concise recommendations over technical environmental measurements.

5. Usability Findings

Finding 1 — AQI interpretation: The numerical AQI was visible, but users wanted a stronger plain-language explanation.

Severity: Medium.

Finding 2 — Alert discoverability: Alerts were reachable but not immediately visible from Home.

Severity: Medium.

Finding 3 — Content hierarchy: Health advice should visually compete less with secondary information and more clearly answer “what should I do?”.

Severity: Medium.

6. Design Improvement Suggestions

1. Add a one-line plain-language status below the AQI number, such as “Moderate — outdoor activity is generally okay.”
2. Add an Alerts shortcut to the Home screen.
3. Increase hierarchy for the recommended action using a dedicated recommendation card.
4. Keep advanced measurements behind a secondary “Details” action.
5. Repeat the usability test after implementation and compare task success/time against this baseline.

7. Iterative Improvement Plan

Before: users must infer what the AQI number means and navigate to Alerts through Forecast.

Change: provide an explicit recommendation and direct alert access.

After: retest the same six tasks and measure whether completion becomes faster and more consistent.

8. Final Conclusion

The usability evaluation shows that the core AirSense AI flow is understandable, while discoverability and information hierarchy are the main areas for improvement. The proposed changes are small, measurable, and suitable for an iterative design cycle.

Included supporting files: test script, observation sheet, demo results CSV, and visual findings sheet.